



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	02
Areas covered	Control Structures, Logical Operators, Loops Statements, Branching Statements

Q1. [Arithmetic + If-Else-If]

Write a program to input marks of 4 Questions. Calculate total, average, and assign grade:

- $\geq 80 \rightarrow A$
- $65-79 \rightarrow B$
- $50-64 \rightarrow C$
- $< 50 \rightarrow F$

Q2. [Unary Operators – Evaluation & Explanation]

Trace and display the output of the following code:

```
int x = 7;
```

```
int y = x++ + ++x + --x + x--;
```

```
System.out.println("x = " + x + ", y = " + y);
```

Explain step-by-step how the final values are obtained.

Q3. [Logical Operators + If]

Write a program to input a number and determine whether it is:

- divisible by both 2 and 7
- divisible by only one of them
- not divisible by either

Q4. [If Statement]

Write a program to input a number and check whether it is a multiple of 5.

Q5. [If-Else]

Input a number and check whether the number is odd or even.

Q6. [If-Else-If Ladder]

Write a program to input electricity units consumed and calculate the bill:

- First 100 units → Rs. 10/unit
- Next 100 → Rs. 15/unit
- Above 200 → Rs. 20/unit

Q7. [Switch Case]

Write a program that accepts a number (1–7) and displays whether the day is:

- Weekday
- Weekend

Q8. [For Loop + Pattern]

Print the following pattern using nested for loops:

```
*
**
***
****
*****
```

Q9. [While Loop]

Write a program to input numbers continuously and calculate the sum of odd numbers only. Stop when the user enters 0.

Q10. [Do-While + Break + Continue]

Design a simple menu-driven system:

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Requirements:

- Initial balance = 5000
- Prevent withdrawal if balance < amount
- Use continue to skip invalid inputs
- Use break to exit the system